

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

WORK EXPERIENCE

PERFOMAX <i>AI Advisor</i>	2026–present
CONSTRUCTOR GROUP <i>Machine Learning Consultant</i>	2021–present
NATIONAL UNIVERSITY OF SINGAPORE <i>Postdoc under Kostya Novoselov</i>	2022–present

Leadership

- Program Chair for the [AI4X 2025/6 conferences](#)
- Main organizer of the [ICLR 2025 Workshop on Machine Learning Multiscale Processes](#)

Wyckoff Transformer for generating symmetric crystals [ICML 2025](#) · [GitHub](#)

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Accelerated inference by 4 orders of magnitude, to 0.05 GPU ms per structure
- Implemented the model in PyTorch; production-ready package published on [PyPI](#)
- Orchestrated large-scale (10k structures) DFT computations with VASP at an HPC cluster
- Led a team of 6

ML for defects in 2D crystals [npj 2D Mater. Appl.](#) · [npj Comput. Mater.](#) · [patent](#) · [GitHub](#)

- Conceived a novel sparse representation of crystals with defects
- Achieved a 3.7x energy prediction error reduction
- Optimized training and inference: 4x less memory, 8x fewer GPU operations
- Developed code for reproducible parallel running of machine learning experiments
- Led a team of 3

CERN [YANDEX → HSE UNIVERSITY] <i>Researcher</i>	2014–2022
2025 Breakthrough Prize in Fundamental Physics as a member of LHCb.	

Generative models uncertainty estimation [paper](#) · [conference](#) · [code 1](#) · [code 2](#)

- Developed the first methods for estimating uncertainty of conditional GANs.
- Led a team of 3 students.

Generative models for fast simulation [paper 1](#) · [paper 2](#) · [paper 3](#) · [code](#)

- Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
- Achieved approximately 10^5 speed-up compared to the ab-initio simulator.

Machine learning on noisy data [paper 1](#) · [paper 2](#) · [code](#)

- Developed a rigorous way to train ML algorithms on data with the label-noise model common in high-energy physics.

Muon identification at the LHCb experiment at CERN [paper](#) · [poster](#) · [IDAO-2019](#)

- Solved a classification problem over tabular data with noisy labels under timing constraints.
- Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
- Integrated the model into LHCb production, mostly in C++.
- Packaged the work as a data science competition problem for IDAO-2019.

CatBoost aka fighting biases with dynamic boosting [paper](#)

- Set up a distributed system for running experiments for the team with Bash and Python.
- Studied gradient boosting improvements with experiments on toy data.

EDUCATION

HSE University & Sapienza Università di Roma <i>PhD in Computer Science and Physics (Double Degree)</i> Supervisors: Andrey Ustyuzhanin & Barbara Sciascia Thesis: Machine Learning for particle identification in the LHCb detector	2016–2020
Product Management course at Yandex <i>Focused coursework in product strategy and execution.</i>	2018
Yandex School of Data Analysis <i>Master's-level CS programme</i> Algorithms, machine learning, deep learning, and distributed systems.	2013–2015
Moscow Institute of Physics and Technology <i>MS in Physics</i> Optimisation of data processing of the LHCb experiment.	2014–2016
<i>BS in Physics</i> Wave-packet molecular dynamics of electrons in nonideal plasma.	2010–2014
International Junior Science Olympiad (IJSO) Silver medal	Korea, 2008

MENTORSHIP

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- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [2](#)).
 - Student council member from 2013 to 2016, leading several dormitory-scale IT projects.
 - Co-PI of a \$3.4m AI Singapore grant on multiscale machine learning.

TEACHING & OUTREACH

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- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
 - Taught at the Machine Learning in High Energy Physics Summer School in [2015](#), [2016](#), [2017](#), [2018](#), [2019](#), [2020](#), [2021](#), and [2023](#).
 - Lecturer in the [award-winning](#) Coursera Advanced Machine Learning [specialisation](#)
 - Taught and developed an introductory data analysis [course](#) at HSE.
 - Delivered more than 20 [conference talks](#).

SERVICE

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- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology, Digital Discovery
 - Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

SKILLS

- **ML & AI** – Structured, non-text, domain-specific data
- **Python** – Research coding
- **PyTorch** – Deep learning
- **C++** – Production
- **Linux** – Administration
- **Research Writing & Speaking**
- **Physics** – Deep domain understanding
- **Leadership & Mentorship**

SELECTED PUBLICATIONS

1. **Kazeev, Nikita**, and Ian Babich. [Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments](#). Forecasting as a New Frontier of Intelligence Workshop at ICML 2026
2. **Kazeev, Nikita**, and Phan Bui Nhat Huyen. [Position: Align AI to Our Aspirations, Not Our Flaws](#). Pluralistic Alignment Workshop at ICML 2026
3. Kazeev, Nikita and Ustyuzhanin, Andrey. [The Geometry of Reasoning Failure: Predicting Agent Errors from Semantic Scale Trajectories](#). Statistical Frameworks for Uncertainty in Agentic Systems Workshop at ICML 2026
4. Artem Dembitskiy, Shuya Yamazaki, Artem Maevskiy, **Nikita Kazeev**, Roman A. Eremin, Semen Bundenyy, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. [Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials](#). AI for Science Workshop at ICML 2026
5. Nian Liu, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Yuwei Zeng, Ryoji Kubo, Pengru Huang, Thomas Laurent, Yann LeCun, Kostya S. Novoselov, and Xavier Bresson. [Crys-JEPA: Accelerating Crystal Discovery via Embedding Screening and Generative Refinement](#). arXiv preprint arXiv:2605.14759, May 2026.
6. Alexandre Duval, Siddharth Betala, Samuel P. Gleason, Andy Xu, Georgia Channing, Daniel Levy, Ali Ramlaoui, Clémentine Fourrier, Chaitanya K. Joshi, **Nikita Kazeev**, Sékou-Oumar Kaba, Félix Therrien, Alex Hernández-García, Rocío Mercado, N M Anoop Krishnan. [LeMat-GenBench: Bridging the gap between crystal generation and materials discovery](#). AI for Accelerated Materials Design - NeurIPS 2025 / arXiv preprint arXiv:2512.04562, October 2025.
7. **Kazeev, Nikita**, et al. [WyckoffTransformer: Generation of Symmetric Crystals](#). ICML 2025, May 2025.
8. **Kazeev, N.**, Al-Maeeni, A.R., Romanov, I. et al. [Sparse representation for machine learning the properties of defects in 2D materials](#). npj Comput Mater 9, 113 (2023).
9. M. Borisyak and **N. Kazeev**. [Machine Learning on data with sPlot background subtraction](#). Journal of Instrumentation 14.08 (2019). **Alphabetic order, I'm the corresponding author.**
10. Derkach, D., **Kazeev, N.**, Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). [Cherenkov detectors fast simulation using neural networks](#). *Nuclear Instruments and Methods in Physics Research Section A*. **Alphabetic order, I'm the corresponding author.**